

Ryan Feng Lin (he/him/his)

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Education

University of Washington, Seattle, WA 01/2021 – 08/2026
Ph.D., Industrial and Systems Engineering
Advisor: Shuai Huang
Dissertation: Trustworthy Personalized Machine Learning Systems under User-System Interactions

University of Science and Technology of China, Hefei, China 09/2017 – 06/2020
M.S., Computer Science
Thesis: Learning with Collaborative Local Disturbance for 3D Skeleton-based Human Action Recognition

University of Science and Technology of China, Hefei, China 09/2013 – 06/2017
B.S., Statistics
B.S., Computer Science (Double degree)

Employment

Mississippi State University, Starkville, MS 08/2026 – Present
Assistant Professor, Industrial and Systems Engineering

Meta Platforms Inc., Bellevue, WA 09/2024 – 09/2025
Visiting Researcher

Mayo Clinic, Rochester, Minnesota 06/2022 – 09/2022
Biostatistician Intern

Honors, Awards, and Fellowships

Curtis Reusser Endowment Travel Award in UW ISE 05/2026

Winner in IISE DAIS Best Student Paper Competition, for the paper “Team, Then Trim: An Assembly-Line LLM Framework for High-Quality Tabular Data Generation” 03/2026

Rising Star Fellow at the 2nd IOE-ISyE-MS&E Rising Stars Workshop 03/2026

Runner-up in INFORMS QSR Best Paper Competition, for the paper “CrowdLLM: A Synthetic Crowd of LLM-based Decision-Makers Augmented with Generative Models” 09/2025

Meta AI Mentorship Fellowship 09/2024 – 09/2025

Finalist in INFORMS QSR Best Paper Competition, for the paper “Fair Collaborative Learning (FairCL): A Method to Improve Fairness amid Personalization” 10/2023

Research Interests

My research lies at the intersection of AI, statistics, and quality science. My work aims to bridge human, knowledge, and system perspectives to advance quality and trustworthy decision-making.

Methodology: machine learning and human-centered AI, model personalization, uncertainty quantification, experimental design, and data-driven optimization.

Applications: healthcare and medicine, transportation, and marketing.

Publications

In this list below, I indicate with: [†] co-first author (★) graduate students I mentor as co-authors
For peer-reviewed conferences, acceptance rates are shown whenever available.

Refereed Journal Papers

- [1] Kehua Chen, **Ryan F. Lin**, Shuchen Zhang and Yinhai Wang. “Deep Fictitious Play-Based Potential Differential Games for Learning Human-Like Interaction at Unsignalized Intersections.” *IEEE Transactions on Intelligent Transportation Systems*, 2026.
- [2] Congjing Zhang[†](★), **Ryan F. Lin**[†], Ruoxuan Bao, Houtao Deng and Shuai Huang. “Team-then-Trim (T2): An Assembly-Line LLM Framework for Synthetic Data Generation.” Under review at *IJSE Transactions*. (Winner of IJSE DAIS Best Student Paper Competition, 2026)
- [3] **Ryan F. Lin**[†], Keyu Tian[†], Congjing Zhang(★), Hanming Zheng, Li Zeng, and Shuai Huang. “CrowdLLM: Building LLM-Based Digital Populations Augmented with Generative Models.” *INFORMS Journal on Data Science*, 2026. (Runner-up of INFORMS QSR Best Paper Competition, 2025)
- [4] Congjing Zhang[†](★), **Ryan F. Lin**[†], Xinyi Zhao, Pei Guo, Wei Li, Lin Chen, Chaoyue Zhao, and Shuai Huang. “ALARM: Automated LLM-Based Anomaly Detection in SmaRt-Home Monitoring with Uncertainty Quantification.” *INFORMS Journal on Data Science*, 2026.
- [5] **Ryan F. Lin**, Chaoyue Zhao, Xiaoning Qian, Kendra Vehik, and Shuai Huang. “Fair Collaborative Learning (FairCL): A Method to Improve Fairness amid Personalization.” *INFORMS Journal on Data Science*, 4(1), pp.67-84, 2025. (Finalist of INFORMS QSR Best Paper Competition, 2023)
- [6] **Ryan F. Lin**, Xiaoning Qian, Bobak Mortazavi, Zhangyang Wang, Shuai Huang, and Cynthia Chen. “Modeling User Choice Behavior under Data Corruption: Robust Learning of the Latent Decision Threshold Model.” *IJSE Transactions*, 56(12), pp. 1307-1320, 2024.

Peer-Reviewed Conference Papers

- [7] Yuantao Wei[†](★), Huiling Liao[†], **Ryan F. Lin**, Congjing Zhang, Xiaoning Qian and Shuai Huang, “Bayesian DAG-IRT: Aggregating Expert Judgments for Causal Graph Recovery.” Under review at *the 33rd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.
 - Average acceptance rate around 25.3% in the past 5 years.
- [8] Chunli Peng[†], **Ryan Feng Lin**[†], Siyuan Xu, Wei-Chien Kao, Congjing Zhang, Yuantao Wei, Jundi Liu, Huiling Liao, Shuai Huang and Xiaoning Qian, “TwinPlan: Evidence-Supported Personalized Health Planning via LLM Agents and Digital-Twin Simulation.” Under review at *the 33rd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.
- [9] **Ryan F. Lin**[†], Yuantao Wei[†], Huiling Liao, Xiaoning Qian, and Shuai Huang. “Position: Causal Learning Should Embrace the Wisdom of the Crowd.” Under review at *the 40th Annual Conference on Neural Information Processing Systems (NeurIPS)*.
 - Average acceptance rate around 25.3% in the past 5 years.
- [10] Yuantao Wei[†](★), Huiling Liao[†], **Ryan F. Lin**[†], and Shuai Huang. “Mixture-of-Chains: Learning Causal Graphs from Human Knowledge.” Under review at *the 40th Annual Conference on Neural Information Processing Systems (NeurIPS)*.
- [11] Congjing Zhang[†](★), **Ryan F. Lin**[†], Noelle So, Meera Patel, Mingqian Li, Huiling Liao, Lin Chen, Xiaoning Qian, and Shuai Huang. “LIFT: Agentic LLMs for Lifecycle-Based Workflow Testing in Healthcare Risk Prediction.” Under review at *the 40th Annual Conference on Neural Information Processing Systems (NeurIPS)*.
- [12] Yunkai Zhang, Qiang Zhang, Diji Yang, **Ryan F. Lin**, Ruizhong Qiu, Benyu Zhang, Hanchao Yu, Jason Liu, Yinglong Xia, Zhuokai Zhao, Lizhu Zhang, Xiangjun Fan, Zhuoran Yu, Abhishek Kumar, and Zeyu Zheng. “Guiding Generative Recommender Systems via Structured Human Priors with Multi-head Decoding.” Accepted by *the ACM Web Conference (WWW)*, 2026.
 - Acceptance rate 20.1% in the year of submission.
 - Oral presentation (Top 9.4%)

- [13] William Yang, **Ryan F. Lin**, Chaoyue Zhao, and Shuai Huang. “A Rawlsian Mixed Integer Programming Approach for Fair Classification.” *Refereed Proceedings of the 2026 INFORMS Optimization Society Conference*.
- [14] Junyuan Hong, Huanhuan Chen, and **Ryan F. Lin**. “Disturbance Grassmann Kernels for Subspace-Based Learning.” *Proceedings of the 24th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1521-1530, 2018.

– Acceptance rate 18.4% in the year of submission.

Papers in Preparation (full preprints expected shortly)

- [15] **Ryan F. Lin**, Hanming Zheng, Ji Liu, Li Zeng and Shuai Huang. “Learning to Collect: A Two-Stage Data Collection Framework for Data-Efficient Personalization.” To be submitted to *Technometrics*.
- [16] **Ryan F. Lin**, Xiaoning Qian, and Shuai Huang. “Relieving the Myopia: Bayesian Active Learning by Mean Objective Cost of Uncertainty with Confidence Ascending.” To be submitted to *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- [17] Jundi Liu, Chunli Peng, **Ryan F. Lin**, Huiling Liao, Siyuan Xu, Shuai Huang and Xiaoning Qian, “A Gamified Platform for Crowdsourcing Decision-Making in Health Management over Digital-Twin-Augmented Health Assistants.” To be submitted to *ICLR 2027*.
- [18] Siyuan Xu, Jundi Liu, Chunli Peng, **Ryan F. Lin**, Congjing Zhang(★), Yuantao Wei(★), Shuai Huang and Xiaoning Qian, “Digital-Twin-Augmented Health Assistants for Wearable Data: Personalized Modeling and Verifiable Prediction.” To be submitted to *Nature Health*.
- [19] Hanming Zheng, **Ryan F. Lin**, Shuai Huang and Li Zeng. “A Unified Framework for Crowdsourcing Quality Control: Identifiability and Estimation of Worker-Granularity Mixture Models.” To be submitted to *INFORMS Journal on Data Science*.

Teaching

Mississippi State University

IE 4933/6933 Information System in Industrial Engineering 08/2026 – Present

University of Washington

IND E 315 Probability and Statistics for Engineers 06/2026 – 08/2026

IND E 412 Integer & Dynamic Programming 03/2024 – 06/2024

IND E 427&527 Data Analytics for Systems Engineering, *Teaching Assistant* 09/2025 – 12/2025

IND E 310 Linear & Network Programming, *Teaching Assistant* 09/2023 – 12/2023

Student Mentoring

Yuantao Wei, *Master’s student in ISE* 05/2025 – 08/2026

Topic: Causal inference, Online learning

– Co-supervise master’s thesis

Congjing Zhang, *PhD student in ISE* 08/2023 – 08/2026

Topic: Large Language Models for Decision-making

– Formulation of research projects; Cultivation of analytical skills (e.g., matrix factorization for uncertainty quantification); Manuscript writing

Grant Writing Participation

National Science Foundation (NSF)

- [1] Collaborative Research: SCH: VR-Assisted Digital Twin of Nurse Fatigue

(PI: Dr. Shuai Huang; pending)

Contribution: Help define the scope and write the complete Task 2 (Hierarchical Collaborative Learning) & Task 4 (M-Optimal Bandit).

- [2] Collaborative Research: III: AI-assisted Knowledge Discovery of Complex Disease
(PI: Dr. Shuai Huang; pending)
Contribution: Help define the scope and write the complete Task 2 (Uncertainty Quantification).

Intelligence Advanced Research Projects Activity (IARPA)

- [3] Translating Fine-grained Human Trajectories into Intelligence by Meaning-making and Reference Matching
(PI: Dr. Shuai Huang; unfunded)
Contribution: Help formulate Thrust 5 (GAN-based Anomaly Detection).

Meta Platforms Inc.

- [4] Optimal Decision in Recommendation (PI: Dr. Shuai Huang; funded)
Contribution: Lead the proposal writing.

Amazon.com Inc.

- [5] A Digital Twin Framework for Smart Pricing (PI: Dr. Shuai Huang; funded)
Contribution: Help formulate Task 1 (Personalized User Choice Behavior Modeling).

Talks & Presentations

- [1] (poster) “Trustworthy and Personalized AI Systems and Digital Populations,” *IOE-ISyE-MS&E Rising Stars Workshop*, 2026, Ann Arbor, MI.
- [2] “CrowdLLM: A Synthetic Crowd of LLM-based Decision-Makers Augmented with Generative Models,” *INFORMS Annual Meeting*, 2025, Atlanta, GA.
- [3] “Learning to Collect: A Two-Stage Data Collection Framework for Data-Efficient Personalization,” *INFORMS Annual Meeting*, 2025, Atlanta, GA.
- [4] “Bayesian Active Learning by Confidence Gradient-based Mean Objective Cost of Uncertainty,” *INFORMS Annual Meeting*, 2025, Atlanta, GA.
- [5] “Create a Digital Population at Scale with Large Language Models (LLMs) and Generative AI through the CrowdLLM Framework,” *Graduate Seminar, Department of Industrial and Systems Engineering, University of Washington*, 2025, Seattle, WA.
- [6] “Relieving the Myopia: Bayesian Active Learning by Confidence Gradient Mean Objective Cost of Uncertainty,” *Quality and Productivity Research Conference (QPRC)*, 2025, Seattle, WA.
- [7] “Benchmarking Fairness of Genetic Risk Score Models for Early-stage Prediction of Type 1 Diabetes from the TEDDY study,” *INFORMS Annual Meeting*, 2024, Seattle, WA.
- [8] “Fairness Amid Personalization: Battling Disparity through Fair Collaborative Learning (FairCL),” *Graduate Seminar of Department of Industrial and Systems Engineering, University of Washington*, 2024, Seattle, WA.
- [9] (poster) “Genetic Risk Score for Type 1 Diabetes across Ethnic Populations via Multitask Learning,” *American Diabetes Association (ADA) Scientific Sessions*, 2024, Orlando, FL.
- [10] “Fair Collaborative Learning (FairCL): A Method to Improve Fairness amid Personalization,” *INFORMS Annual Meeting*, 2023, Phoenix, AZ.

Conference Organization

- [1] *Session Chair*, “Integration of Human, Knowledge and Systems for Quality and Personalized Decision-making,” *INFORMS Annual Meeting*, 2025.
- [2] *Session Co-Chair*, “Data-Driven vs. Rule-Based: Bridging the Gap for Real-World Applications,” *INFORMS Annual Meeting*, 2025.
- [3] *Session Chair*, “Integration of Human, Knowledge and Systems for Quality,” *INFORMS Annual Meeting*, 2024.
- [4] *General Chair*, *China R Conference* (Hefei), 2017.

- [5] *Program Committee Member*, International Workshop on Nature Inspired Computation and Applications, Hefei, 2017.

Reviewer

Journal

IEEE Transactions on Automation Science and Engineering
INFORMS Journal on Data Science
IISE Transactions on Healthcare Systems Engineering
IEEE Transactions on Artificial Intelligence
IEEE Transactions on Neural Networks and Learning Systems
IEEE Transactions on Emerging Topics in Computing
Computers & Industrial Engineering

Conference

ISMB/ECCB 2021
IEEE Intelligent Vehicles Symposium 2026
NeurIPS 2026

Membership

Institute for Operations Research and the Management Sciences (INFORMS)
Quality, Statistics, and Reliability (QSR) at INFORMS
Data Mining (DM) at INFORMS
Institute of Industrial and Systems Engineers (IISE)
American Statistical Association (ASA)
Institute of Electrical and Electronics Engineers (IEEE)
Association for Computing Machinery (ACM)